Group Project Question Menu: Q1-Q16, or your own

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Not sure which to pick? **5**

Course: New Media Technology and Analytics (20001920076) ·**Fall 2026** ·Group project ·*Governing documents:* `project_brief.md`, `project_rubric.md`.

How claiming works. Claim your topic by email: send your team's **top-2 picks** to `huwei72@tongji.edu.cn` (cc `2510096@tongji.edu.cn`, the teaching assistant; subject `NMTA: topic claim`, with your team number and both picks) by **Wed 2026-10-14 23:59**. Menu questions are **first come, first served: one team per question, no duplicates**; confirmed topics are posted Week 5 (Thu 2026-10-15). You may instead propose **your own question** (Q17), which needs instructor confirmation before it is yours.

Each question below names a business framing, the tables it uses, its method family, and a theory lens with a Monday-morning hint. Use them as launch pads, not fences. A good team sharpens the question around what the data actually shows.

Each question ends with an optional **Stretch** line: a robustness check or a second angle for a team of six or seven that wants more depth. Stretch work is never required. Marks follow how well you execute and justify your analysis, not how much you add (brief, section 7).

The menu (16 + 1)

At a glance. Every question leads with one method family from Weeks 3 to 8. Q9, Q10, Q11, Q13 and Q16 are set in the livestreaming marketing of Week 9.

Method family	Questions
Association rules (Week 3)	Q1, Q15
Classification (Week 4)	Q3, Q9, Q10
Regression (Week 5)	Q2, Q8
Clustering (Week 6)	Q4, Q11, Q12
Text analysis (Week 7)	Q5, Q6 (feeding regression), Q16
Social network analysis (Week 8)	Q7, Q13, Q14

Q1 ·Which stock keeping units (SKUs) are repeatedly bought together?

Business framing: the founder wants bundles and a smarter stream sequence: which products does the market already treat as a set? **Tables used:** `lb_order_lines`. **Method family:** association rules (Week 3). **Theory lens + Monday morning:** cross-selling inside the marketing mix (Product x Price); the manager pins one bundle and one "frequently bought together" combo in the next stream. **Stretch (optional):** check whether your top rules hold in both livestream and store-shelf orders, and in both halves of the quarter.

Q2 ·Do mid-tier streamers outsell top-tier streamers per viewer?

Business framing: top-tier hosts cost more and draw bigger rooms, but does their output per viewer justify the booking fee? **Tables used:** `lb_streams`. **Method family:** regression / group comparison (Week 5). **Theory lens + Monday morning:** parasocial trust versus reach in key opinion leader (KOL) economics; the manager sets next month's streamer booking mix by output per viewer, not by follower tier. **Stretch (optional):** add the week of the quarter and event nights as controls, and check whether the gap between the tiers survives.

Q3 ·Who are the high-value users, and can we predict them?

Business framing: the top 20% of buyers carry the quarter; what do they look like on arrival, and can we flag the next ones early? **Tables used:** `lb_users`, optionally features built from `lb_order_lines`. **Method family:** classification (Week 4). **Theory lens + Monday morning:** customer lifetime value and the Pareto principle; the manager targets lookalikes of high-value users with retention offers instead of blanket coupons. **Stretch (optional):** build every feature from each buyer's first 30 days only, so the model flags high-value users early instead of describing them after the fact.

Q4 ·What user segments does the store actually serve?

Business framing: "our customer" is nobody: find the groups hiding in a quarter of orders and activity, and name them so a category manager would recognize them. **Tables used:** `lb_users`, features built from `lb_order_lines`. **Method family:** clustering (Week 6). **Theory lens + Monday morning:** segmentation-targeting-positioning; the manager picks one segment to serve differently and says what changes for it. **Stretch (optional):** show that the segments survive a change of scaling or of k (the number of clusters), and profile each segment's arrival channel and comments.

Q5 ·What do satisfied and dissatisfied customers talk about?

Business framing: ten thousand stream comments are free-text market research nobody has read: what themes separate the happy from the unhappy? **Tables used:** `lb_comments`, optionally joined to `lb_streams` for context. **Method family:** text analysis and natural language processing (NLP), Week 7. **Theory lens + Monday morning:** voice of the customer and expectation-confirmation; the manager fixes the top complaint theme before the next campaign night. **Stretch (optional):** build your own keyword themes, compare them with the platform's sentiment labels, and explain where the two disagree.

Q6 ·Does comment sentiment predict repurchase?

Business framing: the comment feed feels like an early-warning system: does what users say during streams actually track whether they come back to buy? **Tables used:** `lb_comments` joined to `lb_users` (and `lb_streams`). **Method family:** text analysis (Week 7) as the lead family, feeding regression (Week 5), the

menu's one deliberate two-family question. **Theory lens + Monday morning:** the satisfaction-loyalty link and word of mouth; the manager decides whether a sentiment key performance indicator (KPI) earns a permanent place on the weekly dashboard. **Stretch (optional):** make the timing run the right way: measure sentiment before the purchases you predict, never after them.

Q7 ·Who are the key nodes in the share/follow network?

Business framing: shares carry the stream beyond its own room: which users, streamers, or the store account hold the graph together, and where would a seeding budget travel furthest? **Tables used:** `lb_follows`, optionally `lb_users`. **Method family:** social network analysis (Week 8). **Theory lens + Monday morning:** network effects and the strength of weak ties; the manager chooses which nodes to seed with share incentives. **Stretch (optional):** compare the top nodes under two or three centrality measures, and check whether the key nodes are also valuable customers.

Q8 ·Were the event nights worth it?

Business framing: platform and brand campaign nights cost fees and discounts: did event-night streams actually outperform, or would those sessions have sold anyway? **Tables used:** `lb_streams`, optionally `lb_order_lines`. **Method family:** regression with causal intuition (Week 5). **Theory lens + Monday morning:** promotional lift and marketing return on investment (ROI); the manager renews or skips the next campaign night based on the lift per event, not the headline gross merchandise value (GMV). **Stretch (optional):** compare each event night with ordinary nights of the same host tier and a similar week, or add those controls to the regression.

Q9 ·Can the first half hour of a stream tell a strong night from a weak one?

Business framing: by minute 30 the stream team can still change the night: add a flash coupon, move a bestseller forward, or extend the session. Using only what is known by then (the first 30 minutes of comments and the facts fixed before the stream began), can the team tell a strong night from a weak one? Define "strong" before you model. **Tables used:** `lb_comments` (comments posted in the first 30 minutes), `lb_streams` (the outcome, plus the host, tier, weekday, start hour and event flag known in advance; session totals such as `viewers` arrive only at the end). **Method family:** classification (Week 4), applied to the real-time engagement analytics of Week 9. **Theory lens + Monday morning:** leading indicators and the timing of engagement (when a crowd reacts, not only how much); the manager adopts a minute-30 rule that says when to launch the backup offer, and states how often the rule will fire wrongly. **Stretch (optional):** compare your rule with one that uses the pre-stream facts alone, and test both on the last month after training on the first two.

Q10 ·Who should receive a Double 11 pre-sale invitation?

Business framing: at the end of October the store sends a limited number of pre-sale invitations for Double 11 (Singles' Day, the shopping festival on November 11). Using September and October behavior only, which users will buy in November, and how far down the list should the invitations go? **Tables used:** `lb_users` and `lb_order_lines`, split at 2026-11-01 (features from September and October, the outcome from November); optionally `lb_comments` and `lb_follows` for engagement features. **Method family:** classification (Week 4), set in the campaign calendar of Week 9. **Theory lens + Monday morning:** the customer lifecycle and recency, frequency and monetary value (RFM) analysis; the manager sends invitations down the ranked list and stops where the invitations likely to be wasted outweigh the extra November buyers they reach. **Stretch (optional):** compare the model with a rule of thumb such as "two or more orders since September", and show how many November buyers each invitation budget captures (a gains chart).

Q11 ·What kinds of stream nights do we actually run?

Business framing: on the calendar, 78 sessions look alike; in the data, a big-audience night and a small, efficient night are different products with different costs. Find the session types hiding in the quarter and decide which type each slot of next month's calendar should carry. **Tables used:** `lb_streams`, optionally with per-session features built from `lb_comments` and `lb_order_lines` (comments per viewer, basket size, discount depth). **Method family:** clustering (Week 6), applied to the streamer economics of Week 9. **Theory lens + Monday morning:** reach versus conversion as a portfolio choice in streamer economics; the manager plans next month's calendar as an explicit mix of session types, each with its own sales

target. **Stretch (optional):** show that the session types survive a change of scaling or of k, and profile what viewers say and buy in each type.

Q12 ·Which products should lead a stream, and which should ride along?

Business framing: thirty products compete for screen time. Some anchor a basket, some sell mainly as add-ons, and some are cheap entry points. Group the catalog by how each product actually sells, then give each group a role in the stream running order. **Tables used:** `lb_order_lines`, aggregated to one row per product (price level, discount against the store-shelf price, livestream share, how often it shares a basket, repeat buyers); optionally `lb_users`. **Method family:** clustering (Week 6), on 30 products instead of on customers. **Theory lens + Monday morning:** product roles within the marketing mix (Product x Price): anchors, add-ons, and entry products; the manager rewrites the running order so each group appears where its role says it sells best. **Stretch (optional):** confirm the groups with hierarchical clustering and its tree of merges (the dendrogram), and explain every product that changes group.

Q13 ·Do our six hosts reach six audiences, or one?

Business framing: the store pays six hosts. If their followers are largely the same people, booking another host buys conversion, not new reach. Map who follows which host and measure how much of each host's audience no other host reaches. **Tables used:** `lb_follows` (the follow edges to streamers), optionally `lb_users` and `lb_streams`. **Method family:** social network analysis (Week 8) on a two-mode network (users on one side, hosts on the other), applied to the streamer economics of Week 9. **Theory lens + Monday morning:** audience duplication, and reach versus frequency in media planning; the manager decides whether next month's booking budget buys exclusive reach or the best conversion. **Stretch (optional):** compare the overlap you observe with the overlap left after shuffling who follows whom at random, or compare the spend of followers shared by several hosts with that of exclusive followers.

Q14 ·Does a shared stream bring in new buyers?

Business framing: a share reward would pay users each time they send a stream to a friend. Before the store pays, find out whether shares reach new people and whether those people go on to buy. **Tables used:** `lb_follows` (the share edges, with their date and stream), `lb_order_lines` (when each recipient orders), `lb_users`. **Method family:** social network analysis (Week 8): the share network, its reach, and what happens to the people it reaches. **Theory lens + Monday morning:** word of mouth and social contagion, tested against the rival explanation that shares mostly reach people who would have bought anyway; the manager decides whether to reward each share, each referral that converts, or neither. **Stretch (optional):** compare the recipients of the most active sharers with those of occasional sharers, and estimate the highest reward per converted referral the store could pay and still break even.

Q15 ·What should the "you may also like" shelf show, and to whom?

Business framing: after each purchase the mini-program suggests more products. Rules mined from whole purchase histories can end up mirroring the heaviest buyers. Build the shelf from the purchase data and check whom it actually serves. **Tables used:** `lb_order_lines` (single orders and whole customer histories), `lb_users` (order counts, to tell light buyers from heavy ones). **Method family:** association rules (Week 3), mined on customer histories as well as on single orders. **Theory lens + Monday morning:** recommender systems and algorithmic bias (an ethics touchpoint of Week 10); the manager chooses which rules go on the shelf and whether first-time buyers see a different one. **Stretch (optional):** mine the rules separately for light and heavy buyers and compare the two shelves.

Q16 ·What do viewers ask before they buy?

Business framing: many comments are questions, not opinions. A host who answers them before they are asked keeps the stream moving. Find the question themes, when in a stream they arrive, and which of them deserve a pinned answer card. **Tables used:** `lb_comments` (question comments and their timing), optionally `lb_streams` and `lb_order_lines`. **Method family:** text analysis (Week 7), applied to the real-time engagement analytics of Week 9. **Theory lens + Monday morning:** information search in the consumer decision journey; the manager rewrites the host script and the pinned card so the top question themes are answered in the first minutes. **Stretch (optional):** test whether streams with more questions per viewer sell more per viewer, or whether viewers who ask go on to order in the same stream.

Q17 ·Your own question

Business framing: bring a new-media marketing question of your own (from an internship, a brand you follow, a platform phenomenon) framed with the same four parts as Q1-Q16. **Tables used:** the course project dataset, or data you propose and the instructor approves. **Method family:** one method family from Weeks 3-8. **Theory lens + Monday morning:** name the lens and the decision the answer changes. Your question needs instructor confirmation by the claim deadline. **Stretch (optional):** name one check that would make you trust your answer more, and plan time for it. **Data rules:** Dataset: course project dataset (Lumen Beauty full Sep-Nov operations export, five tables) or your own proposal; public, properly-licensed, or realistic synthetic data only; no scraped personal data. Data other than the course project dataset needs a data plan attached to your topic claim: see `project_data_guidebook.md`.

Not sure which to pick?

Choose the question where you can already see the **Monday-morning recommendation** taking shape. A question you only half-understand but genuinely care about will serve you better than a fashionable one that leaves you cold.